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MINI-REVIEW

Non-programmed rehospitalizations after cholecystectomy



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Summary Cholecystectomy is one of the most frequent procedures in digestive surgery. While the operation is generally associated with low rates of morbidity and mortality, frequency of occurrence can vary considerably according to surgical indication, time elapsed between symptom appearance and surgical intervention, anatomical area under treatment, and the experience of the different centers. Rehospitalization after cholecystectomy remains potentially problematic in numerous units, due in part to the ongoing development of day hospital treatment and short-term hospitalization. The objective of this update is to assess not only the rate, causes and risk factors of non-programmed hospitalizations subsequent to cholecystectomy, but also the available ways and means of prevention and management in the patient's best interests.

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Introduction

While overall prevalence of biliary lithiasis is estimated at 10 to 15% in the overall population, it varies pronouncedly from one country to another [1]. Around 30% of patients presenting with biliary lithiasis encounter complications, with annual incidence ranging from 1 to 3% [2]. In the United States, prevalence of symptomatic vesicular lithiasis has doubled over the course of the last three decades [3]. In

Europe, biliary lithiasis represents the main cause of hospital admission due to a gastrointestinal pathology [4].

Cholecystectomy is one of the most frequent procedures in digestive surgery. In France, more than 120,000 patients are operated by year, and the rate of incidence increased from 167 to 183 per 100,000 inhabitants between 2008 and 2014 [5]. Symptomatic biliary lithiasis represents the premier indication for surgery [5]. Over 92% of procedures are performed by coelioscopy [5], and the rate of outpatient care with regard to the number of potentially realizable ambulatory procedures rose from 4 to 50% between 2008 and 2016 [6]. The operation is generally associated with low proportions of morbidity and mortality: 1.6–5.3% and 0.08–0.14% respectively, but pronounced variations may occur according to the indication for surgery, the time

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elapsed between symptom appearance and surgical procedure, the patient's general condition, and the experience of the different centers [7].

Notwithstanding marked improvement of cholecystectomy outcomes over recent years, rehospitalization remains problematic in numerous hospital units, and is often indirectly associated with the accelerated development of outpatient care and short-term hospitalization. The objective of the present updating was to specify the rate of non-programmed rehospitalization subsequent to cholecystectomy and to indicate not only its causes and risk factors, but also how to prevent them and manage in the patient's best interests.

Rehospitalization rates and risk factors

Rates of rehospitalization following cholecystectomy range from 0 to 11%; 7% worldwide according to the different series and definitions (generally 30 or 90 days) [8]. Quite obviously, socio-economic factors characterizing the countries and populations included, not to mention disparate medical practices, affect these rates. In France, rehospitalization rates are not published on a nationwide scale, with the exception of a French multicenter prospective series concerning the randomized ABCAL trial on acute cholecystitis, which reported 7.5% of rehospitalizations after 414 cholecystectomies for acute cholecystitis [9].

The main causes for rehospitalization have been documented in a large-scale register study by Rosero et al. on ambulatory procedures [10]. They are highlighted by residual gallstones in the common bile duct (CBD) (20%), most of which are managed by endoscopy (90%). The other causes encompass pain (16%), surgical site infections (14%), nausea/vomiting (12%) and hemorrhagic complications (8%). Biliary wounds account for 0.008% of reasons for readmission.

Risk factors for rehospitalizations were specifically analyzed in the same register studies [8,10,11]. Some comorbidities including chronic organ failure (chronic renal failure, chronic pulmonary pathologies, chronic hepatopathy) and progressive cancer, as well as demographic characteristics such as male gender, advanced age, ethnic identity (non-Hispanic white) and type of insurance (non-private) have been identified in American cohorts as risk factors for rehospitalization; on the contrary, obesity is not a documented risk factor. Among surgical risk factors, acute cholecystitis is clearly associated with excess morbidity [11] and an increased risk of rehospitalization (7.5% according to the ABCAL trial [9]); cholecystectomy performed during the weekend is likewise associated with an increased risk of rehospitalization, indirectly attesting to the degree of emergency and severity [10,11]. While single-port surgery (vs. four trocars) and ambulatory procedures were not associated with lower readmission rates [8], precise data concerning the impact of surgical approach are few and far between.

Non-specific causes of rehospitalization

Postoperative discomfort

The laparoscopic approach helps to reduce postoperative discomfort, which can associate pain, nausea and vomiting. However, discomfort can be sufficiently substantial to affect

the recovery process, for example by delaying departure from the recovery room, preventing ambulatory management or delaying the patient's discharge, and possibly occasioning rehospitalization [12]. Discomfort prevention is the sole guarantee of a decreased risk of rehospitalization. Moreover, reduction of postoperative pain and, consequently, of analgesic consumption, is important, even in minor abdominal surgery, one reason being that worrisome rates of opioid dependency have been observed following cholecystectomy [13]. In clinical practice, intravenous injection of dexamethasone limits the risk of nausea and vomiting [14]. Perioperatively, systematic infiltration of trocar-related scars by local anesthesia (carried out if possible before incision of the skin) is recommended, especially insofar as it helps to reduce initial analgesic consumption [15]. By means of a local anti-inflammatory effect, infiltration helps to limit the phenomenon of self-induced pain, thereby producing a beneficial effect during a period outlasting persistence of the product at the site of administration. Intraperitoneal instillation of a local anesthetic is frequently proposed, even though its interest has yet to be clearly demonstrated [15]. Mini-laparoscopic cholecystectomy is the most widely evaluated mini-invasive approach, and among selected patients, it appears to decrease the incidence of postoperative pain [16,17]. This approach can be combined with the utilization of reheated and humidified constant pressure CO₂ insufflators, which function at low pressure (9–10 mmHg) and thereby not only attenuate the pain associated with distension of the diaphragm and the abdominal wall, but also avoid the hypothermia and peritoneal irritation contributing to postoperative pain [18]. These multimodal mini-invasive approaches are of particular interest for globally fragile and sickle cell anemia patients [19]. While postoperative pain management necessitates analgesics with levels adapted to pain intensity, it is recommended to concomitantly associate a non-steroidal anti-inflammatory drug helping to limit the consumption of analgesics and to lower the incidence of nausea and vomiting, without increasing the risk of (hemorrhagic...) postoperative complications [14]. Moreover, rigorous postoperative assessment of the patient's overall condition, potentially using a scale such as APAIS, can help to identify patients at risk of developing chronic postoperative pain and who are likely to draw benefit from detailed preoperative information and particularly rigorous peri-operative and postoperative follow-up [14].

More generally, enhanced recovery after surgery (ERAS), including immediate ambulation, same-day refeeding, absence of urinary or abdominal drainage...) is practically systematically applied after laparoscopic cholecystectomy and decreases the risk of rehospitalization due to non-specific causes: fever associated with atelectasis or pulmonary infection, urinary infection, lymphangitis, venous thrombosis... Mechanical or medicine-based thromboprophylaxis (intermittent pneumatic compression) is not recommended in ambulatory surgery or ERAS in a patient at low thromboembolic risk (based on the Caprini score), especially insofar as early deambulation and hydration help to diminish thrombotic risk [20].

Surgical site infection

Just like all other postoperative infections, surgical site infection, at times intraperitoneal or characterized by an abscess at the site of extraction of the surgical speci-

men (generally at incision of the umbilical trocar) can be prevented by strict application of preventive measures during the preoperative, perioperative and postoperative periods: hand hygiene, skin preparation, (non-systematic [21]) antibioprophylaxis and maintained normothermia [21]. More specifically, contamination of the intestinal wall during extraction of the surgical specimen can more often than not be avoided by, before extraction, placing the retrieved specimen in a bag, or else, by protecting the abdominal wall with a skirt [22]. In acute cholecystitis, bacteriological sampling of gallbladder bile is mandatory, especially in case of Grade II severity according to the Tokyo Guidelines and/or a complicated form: necroses, emphysema, perforation [23]. In the postoperative phase, continuation of antibiotic therapy for more than 24 hours is not recommended, except in severe cases (Grade III according to the Tokyo Guidelines and/or in complicated forms), and treatment will be maintained for at least four to seven days in the event of a deep infectious focus following surgery (hepatic abscess) [23]. In case of gallbladder perforation, surgery necessarily involves washing of the surgical site and recovery (if possible) if all the extracted gallstones. Even in the event of perforation, drainage is not systematic [24].

Intestinal wall access is treated by surgical debridement. Management of deep abscess is more complex and may associate radiologic or endoscopic drainage and even, in certain cases, surgical redo (by laparoscopic approach, if possible) according to the positioning of the collection and the available technical platforms [25,26]. Antibiotherapy is generally initiated prior to drainage, adapted according to the results of the bacteriological sampling, and maintained for five to fifteen days [26]. Deep abscesses are generally lacking in precise cause; that said, they may in some cases be entailed by a biliary fistula (see below) or, more rarely, by "lost" intra-abdominal or parietal gallstones, which shall need to be extracted, along with the abscess, in order to avoid possible recurrence. Gallstone ablation may be simple, when the gallstone has migrated in the wall, or more complex, necessitating percutaneous or surgical radiological ablation or more complicated surgery [27].

Hemorrhages

Hemorrhagic complications are relatively rare (2–3%), and may be divided into occasionally severe perioperative hemorrhage (less than 2%) [28] and postoperative hemorrhage, accounting for 1 to 2% of post-cholecystectomy complications, and responsible for 8% of rehospitalizations [8,28], encompassing hematomas of the vascular bed and/or wall. The risk of hemorrhagic complications is compounded by thromboprophylaxis [28,29], which nonetheless does not reduce thromboembolic complications; as a result, it is not systematically recommended. Moreover, the laparoscopic approach is associated with heightened hemorrhagic risk [30], which reflects the degree of surgical difficulty and severity of the vesicular pathology.

Management of postoperative hematomas depends on degree of severity, extent of impact, and the patient's overall condition; it ranges from simple monitoring in a patient in good general condition presenting with a small hematoma without hemodynamic or biological repercussions, to surgical redo in the event of voluminous hematoma responsible for deglobulization. A postoperative hematoma is at risk of surgical site infection and must in all cases be closely monitored. Particular vigilance is required for patients undergoing hemostasis-modifying treatment (anti-

coagulants, antiaggregants), and it begins with painstaking surgical hemostasis of the vascular bed and the trocar insertion points, especially those on the trajectory of the epigastric artery.

Residual gallstone

In a large-scale American registry study, recurrent symptoms associated with a residual gallstone represent the second leading cause of rehospitalization subsequent to cholecystectomy [31]. Persistent gallstones expose the patient to the risk of angiocholitis and/or pancreatitis associated with a long-term increase in morbidity and mortality [32]. While risk of residual gallstone is directly related to the initial lithiasic pathology (presence during the cholecystectomy of one or more gallstones in the CBD, or previous lithiasic migration, angiocholitis or pancreatitis), it may also be impacted by type of preoperative exploration (liver function test, endoscopy or biliary MRI), perioperative management (with or without cholangiography) or, quite obviously, when gallstones are initially present in the CBD, by the means of their management (preoperative or perioperative ERCP or surgical ablation the gallstone).

Any suspected gallstone in the CBD (abnormalities on the liver function test not explained by the clinical picture, or abnormal dilation of the bile ducts observed on ultrasound) necessitates preoperative or perioperative exploration of the CBD, the objective being to avoid leaving behind a residual gallstone. Prior to surgery, biliary MRI or endoscopic ultrasound may be considered. Biliary MRI is advantageous insofar as it does not necessitate general anesthesia and presents high sensitivity and specificity [33]. Endoscopic ultrasound not only presents equivalent sensitivity and specificity [33], but also enables, if necessary, elimination of a non-lithiasic cause of obstruction (ampullae, pancreatic tumor), and possibly even to perform, during general anesthesia, ERCP or extraction of a gallstone. The non-invasive nature of biliary MRI has impelled some teams to have it routinely included in preoperative protocols. While this attitude may potentially reduce the risk of rehospitalization due to residual gallstone, its cost–benefit ratio remains to be determined [34].

The possible interest of systematic perioperative cholangiography in reducing the risk of residual gallstone has been widely debated in the literature. Routine use of this relatively non-invasive test, which is not always simple to apply, has been recommended (or suggested) by several learned societies; that said, the literature is highly heterogeneous [35]. In the largest meta-analysis (440,659 patients), the authors observed no reduction of this risk in the cholangiography group [36]. The interest of systematic perioperative cholangiography in diminishing the risk of MBD injury will be considered below.

How cholecystectomy is performed has a potential impact on risk of residual gallstone, which is according to several reports greater following laparoscopic surgery than after open surgery [32,37,38]. Subtotal cholecystectomy, which may be of interest in the event of difficult dissection during management of chronic cholecystitis, may entail the risk of leaving in place a residual gallstone in gallbladder remnant. Systematic ablation of gallstones during the procedure is consequently of prime importance. Moreover, it has been suggested that leaving a very long (more than 15 mm) cystic duct in place during cholecystectomy would increase the risk of residual gallstone [39]. In practice, it is often dif-

difficult to precisely determine during an operation the exact length of the cystic duct; that said, it is always possible to move a possible gallstone towards the gall bladder lightly compressing the latter.

As for the postoperative phase, no systematic examination has shown proven interest in view of reducing the risk of rehospitalization for residual gallstone. That said, post-operative control of liver function normalization, which may have been perturbed by pedicle inflammation during cholecystitis, after successful migration, or following extraction of a MBD gallstone, is usually inexpensive, and is frequently carried out by surgical teams.

Management of residual gallstones will take into account their position (gallbladder remnants, cystic duct, MBD...), the patient's overall condition and medical history (age, previous Roux-en-Y gastric bypass surgery...) as well as the cause of rehospitalization (abscess on residual vesicular remnants, simple migration, angiocholitis...).

A persistent residual gallstone in vesicular remnants can be responsible for rehospitalization due to abdominal pain, associated or not with sepsis or jaundice [40]. Imagery (ultrasound generally postoperatively associated with abdominal scan) highlights a "cholecystitis" aspect of the remnants. These examinations are often subsequently completed by biliary MRI and/or echographic endoscopy. More often than not, initial management involves antibiotic therapy. If clinical evolution is favorable, one may consider (taking into account the patient's overall condition, the available technical platform, and the center's experience) different options in view of avoiding recurrence [40]: redo surgery some time after the acute episode to remove the residual gallstone, cholecystectomy by laparoscopy or laparotomy (procedure that is not always simple, and must be based on precise assessment of the modalities of the initial operation: experience of the surgeon, descriptions of the difficulties encountered in the operation report, anatomopathological examination of the surgical specimen...), ablation of the residual gallstone by endoscopy (ERCP and Dormia basket extraction or lithotripsy), which presents the advantage of being less invasive and is of particular interest for fragile patients, but is not always practicable.

Management of rehospitalization due to symptomatic residual gallstone in the MBD (simple biliary pain associated with liver function perturbations, angiocholitis or pancreatitis) does not essentially differ from management of any other MBD gallstone. That said, the following three points must be kept in mind. First, the earlier the rehospitalization, the higher the risk of biliary fistula, due largely to increased tension in the MBD and, consequently, of the cystic duct remnants; this should impel the surgical teams to remove the MBD obstacle as rapidly as possible, generally by ERCP. Secondly, any persistent gallstone, particularly in the cystic duct, presents a risk of recurrence and must, if at all possible, be extracted. Third, especially in young patients, symptom recurrence or doubt concerning the imaged presence of intrahepatic gallstones, suggests possible genetic predisposition to biliary gallstone formation (Low-Phospholipid Associated Cholelithiasis) and necessitates orientation of the patient toward an expert center to confirm the presence of intrahepatic gallstones with a "comet tail" aspect, search for a mutation of the ABCB4 gene (inconsistency...), and apply ursodeoxycholic acid treatment.

Bile duct injury

Overall incidence of post-cholecystectomy biliary complications is 1.5% including biliary leaks [41], and bile duct injury (BDI) occurs at a frequency ranging from 0.15% to 0.36% (major injuries), which is higher than the rate of 0.1% to 0.2% reported during the epoch of open cholecystectomy. Cholecystitis remains the main risk factor for BDI during cholecystectomy. BDI is responsible for excess morbidity, long-term excess mortality and consequential costs for healthcare systems, and is also one of the most common causes of litigation against digestive surgeons.

Rehospitalization due to BDI injury is rare, involving less than 0.01% of the cases of rehospitalization after cholecystectomy. These rare cases correspond to injury that has not been identified preoperatively, and it may be related to varied clinical pictures: biloma or choleperitonitis in an open wound (bile duct or bile canal section), jaundice ± angiocholitis in occlusion wound (clip, staples on the main bile duct). Management must necessarily be organized in an expert center and is discussed on a case-by-case basis. A biliary fistula usually necessitates emergency drainage by surgery in cases of choleperitonitis, or percutaneously in cases of biloma. Jaundice does not necessarily entail emergency biliary drainage, except with associated angiocholitis. Of note, biliary dilatation may represent an anatomical factor favorable to subsequent surgical repair. After (possible) drainage and prior to any repair procedure, the topography of the wound must be painstakingly assessed by means of imagery (bili-IRM), and associated vascular lesions responsible for slow extension by an ischemic mechanism of the initial lesions must likewise be evaluated (by angio-TDM). Repair of the wound itself can be endoscopic, surgical, immediately or delayed, and choice of the suitable option depends on several topographic, anatomical and vascular factors, as well as the patient's general state of health. A simple lateral leak may be covered by an endoscopic prosthesis, but resection of the convergence with an associated vascular lesion can entail complex repair by hepaticojejunal anastomosis along multiple ducts and, in some cases, will necessitate associated hepatectomies; lastly but very rarely, liver transplant may be required.

Several surgical societies have issued BDI prevention recommendations [42,43]; here are the main messages.

Diagnostic criteria and preoperative time lapses in cases of acute cholecystitis favoring BDI

Due to possibly associated pedicular inflammation, acute cholecystitis is a major BDI risk factor. The Tokyo recommendations [44,45] classify cholecystitis in the terms of three levels of severity according to clinico-biological presentation. On the basis of this classification and in view of reducing the risk of BDI, it is recommended for Grade I (mild) patients to perform laparoscopic cholecystectomy within seven days (ideally in 72 hours) after symptom onset. As for Grade II (intermediate) patients, emergency laparoscopic cholecystectomy is called for, provided that the patient's general condition is good, and that laparoscopy is technically feasible. When the patient's general state of

health is seriously altered, early biliary drainage or delayed laparoscopic cholecystectomy are recommended. For Grade III (severe) patients, emergency biliary drainage is recommended, except in the event of preserved general state of health and low surgical risk, in which case laparoscopic cholecystectomy can be proposed in specialized centers.

Does there exist preoperative imagery reducing BDI risk?

For the diagnosis of acute cholecystitis without suspected gallstone migration in the main biliary duct (MBD), ultrasound remains the gold standard. In case of suspected associated MBD gallstone, bili-MRI or endoscopic ultrasound $\pm$ ERCP may be considered (according to the context) [43]. Bili-MRI presents several advantages: it is rapid and non-invasive, it shows good sensitivity and specificity in the diagnosis of biliary obstruction (> 95%), and it permits the study of biliary anatomy via reconstruction. While preoperative MRI highlighting of anatomical biliary variations can potentially prevent BDI risk, this has yet to be scientifically demonstrated.

Good operative practices: Meeting the Critical View of Safety (CVS) challenge:

It has now been demonstrated that when obtained perioperatively, CVS status is associated with a decreased risk of BDI [42]. As described by Strasberg et al., CVS must encompass the three following elements: (1) the hepatocytic triangle is not obstructed by adipose or fibrous tissue; (2) the lower third of the gall bladder is separated from the liver, thereby exposing the cystic plate; (3) two and only two structures entering the gall bladder must be visualized [46]. In the context of cholecystitis, this is the keystone of BDI prevention, and any failure to obtain CVS should lead to gradual modifications of surgical strategy: calling upon a colleague for assistance, performance of perioperative cholangiography, choice of a "second option" technique: antegrade or subtotal cholecystectomy, or conversion into laparotomy.

The role of perioperative cholangiography:

Notwithstanding sizable literature but with a very low level of evidence, routine perioperative cholangiography during cholecystectomy is not recommended. On the other hand, in the when biliary anatomy is insufficiently ascertained, it is recommended by the experts as a means of reducing the risk of BDI, particularly in patients with acute cholecystitis [42,43].

Alternative approaches in the event of significant pedicular inflammation

When Calot's triangle is severely inflammatory, or when dissection would entail a significant risk of bile duct injury, the antegrade "dome-down" or "fundus-first" technique is an alternative to the classic retrograde technique. Application requires clear comprehension of the anatomy of the cystic and hilar plates. Dissection must be maintained along the subserosal internal layer so as to avoid any vascular and/or bile duct lesion. If the subserosal internal layer around the Calot triangle cannot be exposed, dissection must not be further pursued, and subtotal cholecystectomy

may be preferred so as to avoid vasculobiliary injury due to surgical "error traps".

Subtotal cholecystectomy consists in opening the gall bladder along its axis up to the collar and removing gallstones in such a way as to clearly identify the cystic duct. Painstaking extraction of all gallstones is important in view of avoiding any and all postoperative collection. Two types of subtotal cholecystectomy have been described according to closure or non-closure:

- in subtotal fenestrating cholecystectomy (no residual bile duct remnant), the collar of the remaining gall bladder remains open. This approach may entail a postoperative biliary fistula, which is generally spontaneously resolved;
- in subtotal reconstituting cholecystectomy (existence of gall bladder remnants), what remains of the gall bladder is closed off by sutures.

Given the persistence of the remnants, this approach can entail the formation of more gallstones and expose the patient to the risk of recurrent cholecystitis necessitating a potentially difficult but relatively rare (less than 5%) redo operation. Performance of subtotal cholecystectomy must be described in the operation report, and the patient has to be informed of the fact that part of his or her gall bladder has remained in place.

When they occur, bile duct injuries are often identified preoperatively [47,48], occasionally permitting direct repair by an experienced hepatobiliary surgeon in the absence of any associated vascular lesion (10–20% of cases); such a lesion can be responsible for secondary ischemic-type biliary lesions, and compromise the repair. In all cases, it is recommended that the surgeon be able to call upon a colleague for assistance in the event of complex cholecystectomy, and, in case of suspected BDI, to solicit the opinion of a specialized team. Indeed, specialized management by a tertiary team equipped with a surgical, radiological and endoscopic technical platform is associated with more satisfactory short and long-term outcomes [42]. In clinical practice, topography of the injury and associated vascular damage have an impact on therapeutical attitude, which brings together specialized endoscopic management and/or early or late surgical repair.

Conclusions

Rehospitalization after cholecystectomy is a frequent occurrence in numerous hospital units worldwide (up to 17%). In France, while rate of rehospitalization has not been closely studied, it has been reported as 7.5% in the specific case of patients operated for acute cholecystitis [9]. The main causes of rehospitalization are residual MBD gallstones (20%), pain (16%), surgical site infections (14%), nausea/vomiting (12%), and hemorrhagic complications (8%). Much dreaded, bile duct injuries represent 0.008% of causes for readmission. Prevention of rehospitalization necessitates rigorous selection of surgical indications (based partially on preoperative imagery), characterized by pre or perioperative BDI exploration in case of doubt on the presence of a gallstone in this area; information given to the patient and his/her kith and kin; pre, peri and postoperative application of protocols on surgical discomfort management and, more generally and if possible, application of protocols on enhanced recovery after surgery (ERAS); performance of surgery in compliance with several rules, including Critical View of Safety certification (or, if necessary, performance of antegrade or subtotal cholecystectomy); calling upon a

colleague in the event of doubt on perioperative bile duct injury and/or soliciting an opinion in a specialized center; ablation of the different vesicular gallstones; obtaining satisfactory hemostasis; non-contamination of the intestinal wall during extraction of the surgical specimen. Rehospitalization management varies pronouncedly according to its causes, which may range from the need for optimized medical treatment of postoperative discomfort inadequately controlled in a home setting to the need for treatment in a specialized center of bile duct injury.

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Keypoints

- Rehospitalization rates after cholecystectomy range worldwide from 0 to 11.7% according to series and definitions. In France, rehospitalization rates are not published on a nationwide scale, with the exception of acute cholecystitis (7.5%).
- The causes of rehospitalization are dominated by residual gallstones in the main bile duct (20%), pain (16%) surgical site infections (14%), nausea/vomiting (12%) and hemorrhagic complications (8%). Bile duct injuries constitute 0.008% of the causes of readmission.
- Among the surgical risk factors, acute cholecystitis is clearly associated with excess morbidity and increased risk of rehospitalization.
- Prevention of rehospitalization calls for:
 - rigorous selection of the indications for surgery, particularly including preoperative or perioperative exploration of the primary gall bladder in cases of suspected gallstone in that area;
 - informing the patient and his/her kith and kin;
 - preoperative, perioperative and postoperative application of the relevant surgical discomfort management protocols and, more generally, if possible, application of enhanced recovery after surgery (ERAS) protocols;
 - performance of an operation in full compliance with several rules: acquisition of critical view of safety assessment (or performance, if necessary, of an antegrade or subtotal cholecystectomy), calling a colleague when experiencing doubt on the need for induction of perioperative bile duct injury or request for advice from a specialized center, ablation of the different gallbladder stones, acquisition of satisfactory hemostasis, non-contamination of the intestinal wall during extraction of the surgical specimen.
- Routine performance during the cholecystectomy of cholangiography in view of preventing the risk of residual gallstone or bile duct injury is not recommended. On the other hand, in case of uncertainty on biliary anatomy, the experts recommend cholangiography, especially in patients with acute cholecystitis.

Disclosure of interest

The authors declare that they have no competing interest.

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